Programming(0)

Sayed Mostafa Modaghegh



fall semester 2025 _ Gharjistan uni

Introduction to Python



Learning Objectives

1) Understand what Python is and why it’s used
2) Learn to write and run a Python program
3) Use print() to display information
4) Store data in variables
5) Perform arithmetic operations
6) Get user input with input()
7) Convert data types (casting)



What is Python?

Python is a high-level, easy-to-learn programming language.

Created by Guido van Rossum in 1991.

Used in:

Web development 🌐

Data Science 📊

Artificial Intelligence 🤖

Automation ⚙️



Why Python?

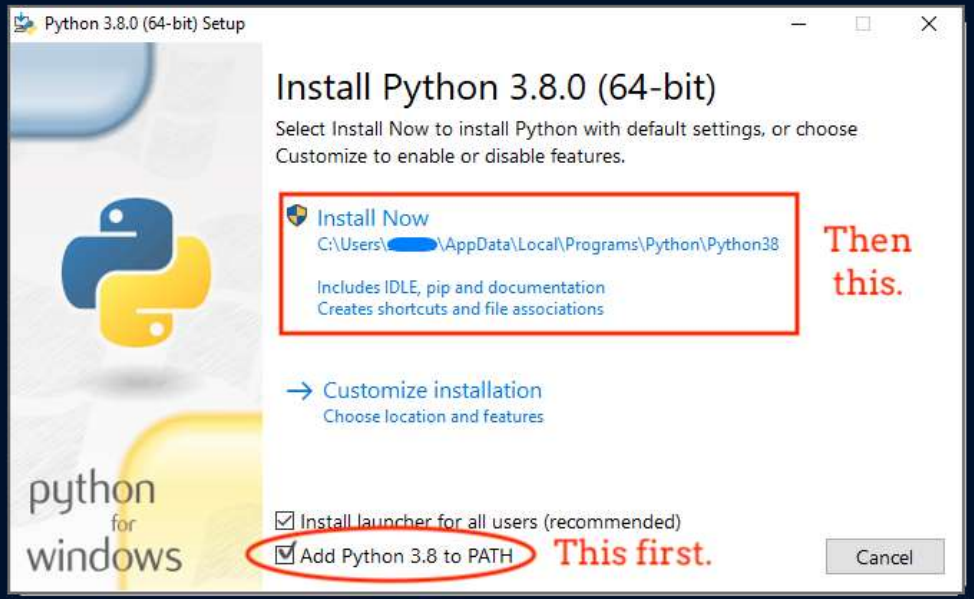
- Simple syntax (like English)
- Huge community and support
- Works on all platforms (Windows, Linux, Mac)
- Great for beginners and professionals





Installing Python

1. Go to python.org
2. Download the latest version
3. Check the box ✅ Add Python to PATH



Where to Write Python Code?

- IDLE (Python's default editor)
- VS Code (recommended IDE)
- PyCharm (advanced IDE)
- Command Line / PowerShell
- Pydroid 3 (for Android users 📱)











First Program

- Create a python file.
- ‘hello.py’ is a python file.
- First code: print()
- print(“Hello World!”)
- In cmd : python hello.py



Variables

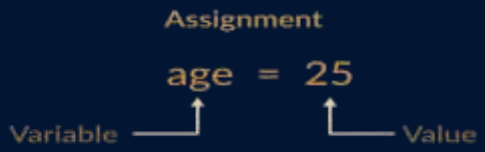


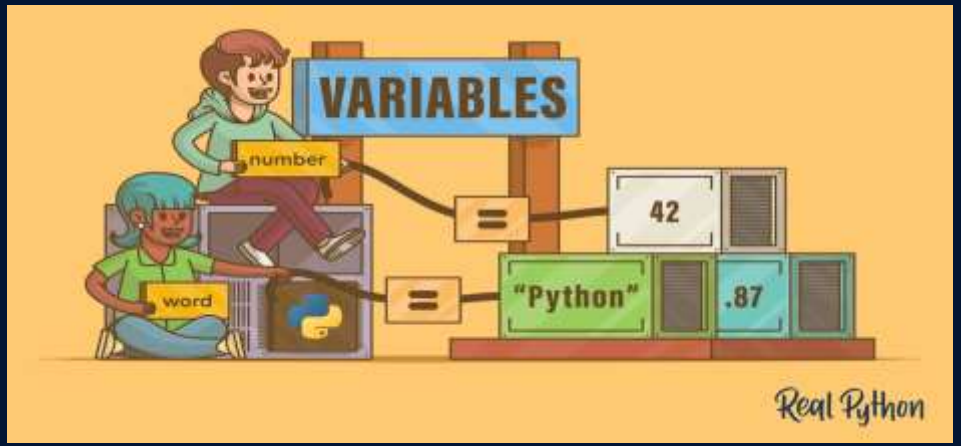
What is a Variable?



Variable = A container that stores data

Think of it like a box with a label 🧰





Variable Naming Rules

- ✅ Must start with a letter or underscore _
- ✅ Can contain letters, numbers, and underscores
- ❌ Cannot start with a number
- ❌ No spaces or special symbols

```
1   my_name = "Ali"
2   age2 = 25
3   _name = "Ali"
4
```



Datatypes

- String
- Integer
- Float




```
1   #Data types:
2   var1 = "Ali"     #string   >>  str
3   var2 = 35        #integer  >>  int
4   var3 = 23.5      #float    >> float
5
```



Arithmetic Operators



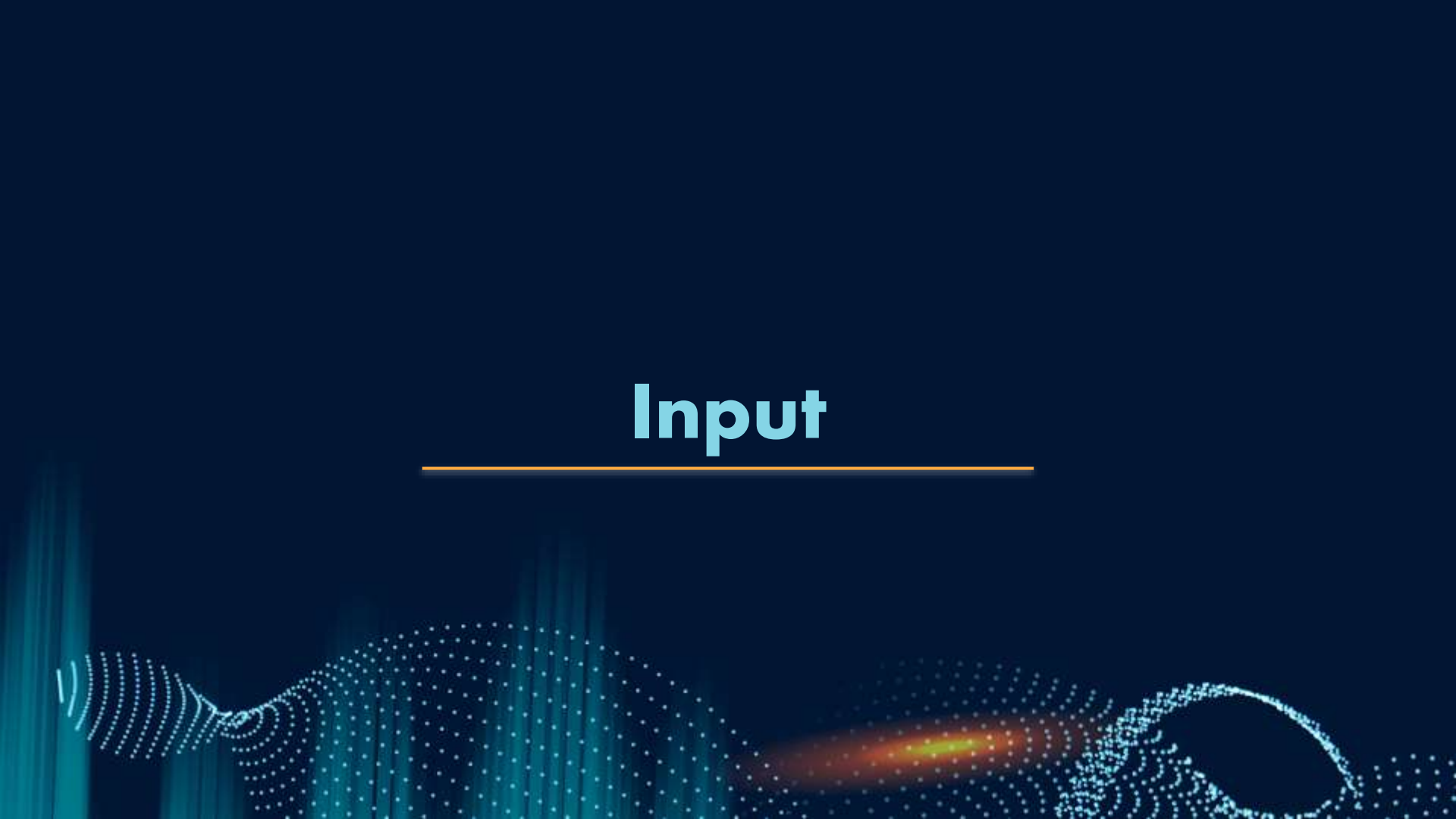
Arithmetic Operators

Arithmetic Operators

Operator	Meaning	Example
+	Addition	4 + 7 ⟶ 11
-	Subtraction	12 - 5 ⟶ 7
*	Multiplication	6 * 6 ⟶ 36
/	Division	30 / 5 ⟶ 6
%	Modulus	10 % 4 ⟶ 2
//	Quotient	18 // 5 ⟶ 3
**	Exponent	3 ** 5 ⟶ 243



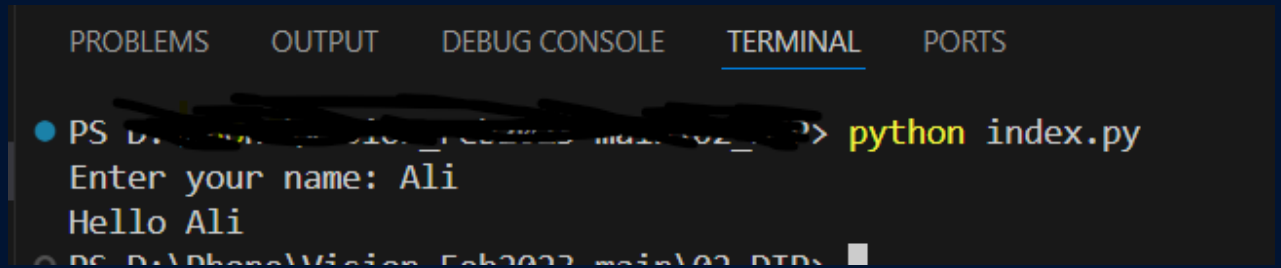
Input



Getting Input from User

```
1   name = input("Enter your name: ")
2   print("Hello", name)
3
```

input() always returns the value as a string.





Thank You

About me:
Sayed Mostafa Modaghegh
AI and Robotics student
Computer vision developer



Sm.mostafa79modaqeq@gmail.com

@mostafa_modaghegh